

Data Science

CSE558

Recap

- Quiz - 1
- Likelihood Estimation

Quiz (Q6)

Expected #tosses until 15Hs and 15Ts.

1. $n > 30$ (Ofcourse)
2. $n = 30$ (too optimistic)
3. $n \geq 30$ (1 or 2)
4. $n = \text{a large number}$ (too pessimistic)
5. $n \geq 60$

Geometric random variable



Overview

- Likelihood Estimation
- Aposteriori Estimation

Likelihood Estimation

Why maximum likelihood and maximum log likelihood are effectively the same?

MLE of Normal Distribution

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are independent random samples from a normal distribution
What is the mean and the variance of the distribution, that maximizes the likelihood of the observed samples.

Suppose for $1 \leq i \leq n$, $-\infty < \mathbf{X}_i < \infty$

$$L = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(\mathbf{X}_i - \mu)^2}$$

$$\log(L) = \frac{-n}{2} (\log(2\pi) + \log(\sigma^2)) - \frac{1}{2\sigma^2} \sum_{i=1}^n (\mathbf{X}_i - \mu)^2$$

$$\frac{\partial \log(L)}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (\mathbf{X}_i - \mu) \qquad \frac{\partial \log(L)}{\partial \sigma^2} = \frac{-n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (\mathbf{X}_i - \mu)^2$$

Question

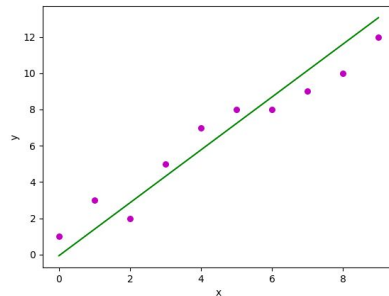
3. Let you get a sample population with their ages.

Age	20	30	40	50	60
Frequency	2	4	6	5	3

What are the parameters of a normal distribution that fits the above sample?

Example

x	0	1	2	3	4	5	6	7	8	9
y	1	3	2	5	7	8	8	9	10	12



$$y = \mathbf{x}^\top \mathbf{w} + \epsilon$$

$$\epsilon | \sigma^2 \sim \mathcal{N}(\epsilon | 0, \sigma^2)$$

$$\Pr(y_n | \mathbf{x}_n, \mathbf{w}, \sigma^2) = \frac{1}{\sigma \sqrt{2\pi}} \exp \left\{ -\frac{1}{2\sigma^2} (y_n - \mathbf{x}_n^\top \mathbf{w})^2 \right\}$$

$$\Pr(\{y_n\}_{n=1}^N | \{\mathbf{x}_n\}_{n=1}^N, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \frac{1}{\sigma \sqrt{2\pi}} \exp \left\{ -\frac{1}{2\sigma^2} (y_n - \mathbf{x}_n^\top \mathbf{w})^2 \right\}$$

$$\log \Pr(\mathbf{y} | \mathbf{X}, \mathbf{w}, \sigma^2) = -\frac{N}{2} \log(2\sigma^2\pi) - \frac{1}{2\sigma^2} (\mathbf{X}\mathbf{w} - \mathbf{y})^\top (\mathbf{X}\mathbf{w} - \mathbf{y})$$

Properties

1. Easy to Compute.
2. Consistent.
3. Efficient.
4. Overfits.
5. MLE need not be unique.

Overview

- Likelihood Estimation
- Aposteriori Estimation

Conditional Probability

Posterior

- $P(H | E)$: Probability that **your friend have the disease** given **he tested positive**
- $P(H | E) = P(E \cap H)/P(E)$

Prior

- $P(H)$: Probability of **having the disease**

Likelihood

- $P(E | H)$: Probability that **he would test positive** given **he had the disease**

Marginal

- $P(E)$: Probability of **testing positive**
- $P(E) = P(H)P(E | H) + P(\sim H)P(E | \sim H)$
- Probability of **having the disease** and **correctly tested positive** +
- Probability of **not having the disease** and **falsely tested positive**

Conditional Probability

Bayes' Rule

$$\begin{aligned}\mathbf{P}(H|E) &= \frac{\mathbf{P}(E|H) \cdot \mathbf{P}(H)}{\mathbf{P}(E)} \\ &= \frac{\mathbf{P}(E|H) \cdot \mathbf{P}(H)}{\mathbf{P}(H) \cdot \mathbf{P}(E|H) + \mathbf{P}(\sim H) \cdot \mathbf{P}(E|\sim H)}\end{aligned}$$